Exploratory Data Analysis (EDA) Report

Diabetes Dataset

This report summarizes the Exploratory Data Analysis (EDA) performed on the diabetes dataset. The goal was to understand the data, check data quality, study relationships between features, and identify useful patterns before machine learning.

1. Dataset Overview

The original dataset contained 100,000 rows and 9 columns. The target variable is 'diabetes': 0 means no diabetes and 1 means diabetes.

Item	Result
Original rows	100,000
Columns	9
Duplicate rows found	3,854
Rows after duplicate removal	96,146

2. Missing Values

The dataset was checked for null or missing values. No important missing-value problem was identified, so no missing-value imputation was required during EDA.

3. Distribution Analysis

Histograms were used to understand the distributions of numerical features such as age, BMI, HbA1c level, and blood glucose level. They helped show where most observations are concentrated and how values are spread.

4. Correlation Analysis

A correlation heatmap was created to study linear relationships between features and diabetes.

Feature	Correlation with diabetes
Blood glucose level	0.42
HbA1c level	0.40
Age	0.26
BMI	0.21
Hypertension	0.20
Heart disease	0.17
Gender	-0.04

Blood glucose and HbA1c had the strongest positive correlations with diabetes. Age, BMI, hypertension, and heart disease had weaker positive relationships. Gender had almost no linear correlation. Correlation shows association, not causation.

5. Outlier Analysis

Boxplots and the IQR method were used to identify outliers in continuous numerical features.

Feature	Total IQR outliers	Direction
Age	0	None
BMI	7,086	1,121 lower; 5,965 upper
HbA1c level	1,315	1,315 upper
Blood glucose level	2,038	2,038 upper

Most detected outliers were on the upper side. The upper IQR limits were 38.505 for BMI, 8.3 for HbA1c, and 247.5 for blood glucose. Maximum values were 95.69, 9.0, and 300 respectively. These observations were not automatically removed because a statistical outlier is not necessarily an error.

6. Outliers and Diabetes

Feature	Outliers	Diabetes status
BMI	5,965 upper	1,466 diabetes=1; 4,499 diabetes=0
HbA1c	1,315 upper	All 1,315 were diabetes=1
Blood glucose	2,038 upper	All 2,038 were diabetes=1

High HbA1c and blood glucose observations were strongly associated with the diabetic class. Removing them only because they are IQR outliers could remove useful information.

7. Duplicate Analysis

Exact duplicate rows were checked using pandas duplicated(). A total of 3,854 duplicate rows were found. After removing duplicates, 96,146 unique rows remained. Removing exact duplicates prevents repeated identical observations from receiving extra weight in analysis or model training.

8. Scatter Plot: Blood Glucose vs BMI

The scatter plot showed that diabetic observations were more concentrated at higher blood glucose levels, while BMI was spread across a wide range for both classes. Blood glucose therefore showed clearer visual separation of the diabetes classes than BMI.

9. Average Blood Glucose by Diabetes Class

The bar plot showed an average blood glucose of about 133 for diabetes=0 and about 194 for diabetes=1. Therefore, the diabetic group had a noticeably higher average blood glucose level. The value 133 is an average, not a diabetes cutoff.

10. Diabetes Percentage by Blood Glucose Range

Glucose range	Diabetes percentage
0–100	0%
121–140	8.41%
141–160	6.95%
181–200	8.51%
201–220	100%
221–240	100%
241–260	100%
261–300	100%

Some ranges were absent because there were no observations in those ranges. In this dataset, every observation above 200 mg/dL (201 and above in the selected bins) was classified as diabetes=1. This is a dataset-specific pattern, not a general medical cutoff.

11. Target Distribution

The target is imbalanced: 91.5% of observations are diabetes=0 and 8.5% are diabetes=1. Therefore, accuracy alone may be misleading when evaluating a machine-learning model.

12. Gender vs Diabetes

Gender	No diabetes	Diabetes
0.0	90.25%	9.75%
1.0	92.38%	7.62%

The diabetes percentages are fairly close between the two gender categories, so no strong difference was observed.

13. Smoking History vs Diabetes

Smoking history	No diabetes	Diabetes
No Info	95.94%	4.06%
Current	89.79%	10.21%
Ever	88.21%	11.79%
Former	83.00%	17.00%
Never	90.47%	9.53%
Not current	89.30%	10.70%

Former smokers had the highest observed diabetes percentage (17%), while the No Info category had the lowest (4.06%). This shows an association in this dataset, but it does not prove that smoking history causes diabetes.

14. Overall Findings

- No important missing-value problem was found.
- 3,854 exact duplicate rows were identified and removed, leaving 96,146 rows.
- Blood glucose and HbA1c had the strongest positive correlations with diabetes.
- Most detected continuous-feature outliers were upper-side outliers.
- All detected upper outliers for HbA1c and blood glucose belonged to diabetes=1.
- The target is imbalanced: 91.5% non-diabetic and 8.5% diabetic.
- The diabetic group had a higher average blood glucose level than the non-diabetic group.
- In the analyzed bins, all observations above 200 mg/dL were classified as diabetes=1.
- Smoking-history categories showed different diabetes percentages, with former smokers highest.

15. Conclusion and Next Steps

The EDA indicates that blood glucose and HbA1c are particularly important features for understanding the diabetes target. The data is also imbalanced, which should be considered during model evaluation. The next stage can include final categorical encoding, feature/target separation, train-test splitting, scaling where appropriate, model training, and evaluation using precision, recall, F1-score, and a confusion matrix.